

Ciguatera: recent advances but the risk remains.

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Ciguatera is an important form of human poisoning caused by the consumption of seafood. The disease is characterised by gastrointestinal, neurological and cardiovascular disturbances. In cases of severe toxicity, paralysis, coma and death may occur. There is no immunity, and the toxins are cumulative. Symptoms may persist for months or years, or recur periodically. The epidemiology of ciguatera is complex and of central importance to the management and future use of marine resources. Ciguatera is an important medical entity in tropical and subtropical Pacific and Indian Ocean regions, and in the tropical Caribbean. As reef fish are increasingly exported to other areas, it has become a world health problem. The disease is under-reported and often misdiagnosed. Lipid-soluble, polyether toxins known as ciguatoxins accumulated in the muscles of certain subtropical and tropical marine finfish cause ciguatera. Ciguatoxins arise from biotransformation in the fish of less polar ciguatoxins (gambiertoxins) produced by *Gambierdiscus toxicus*, a marine dinoflagellate that lives on macroalgae, usually attached to dead coral. The toxins and their metabolites are concentrated in the food chain when carnivorous fish prey on smaller herbivorous fish. Humans are exposed at the end of the food chain. More than 400 species of fish can be vectors of ciguatoxins, but generally only a relatively small number of species are regularly incriminated in ciguatera. Ciguateric fish look, taste and smell normal, and detection of toxins in fish remains a problem. More than 20 precursor gambiertoxins and ciguatoxins have been identified in *G. toxicus* and in herbivorous and carnivorous fish. The toxins become more polar as they undergo oxidative metabolism and pass up the food chain. The main Pacific ciguatoxin (P-CTX-1) causes ciguatera at levels=0.1 microg/kg in the flesh of carnivorous fish. The main Caribbean ciguatoxin (C-CTX-1) is less polar and 10-fold less toxic than P-CTX-1. Ciguatoxins activate sodium ion (Na⁺) channels, causing cell membrane excitability and instability. Worldwide coral bleaching is now well documented, and there is a strong association between global warming and the bleaching and death of coral. This, together with natural environmental factors such as earthquakes and hurricanes, and man-made factors such as tourism, dock construction, sewage and eutrophication, may create more favourable environments for *G. toxicus*. While low levels of *G. toxicus* are found throughout tropical and subtropical waters, the presence of bloom numbers is unpredictable and patchy. Only certain genetic strains produce ciguatoxins, and environmental triggers for increasing toxin production are unknown.